DSCI-510 Final Project

The Evolution of Popular Over Two Decades

Patterns and Popularity

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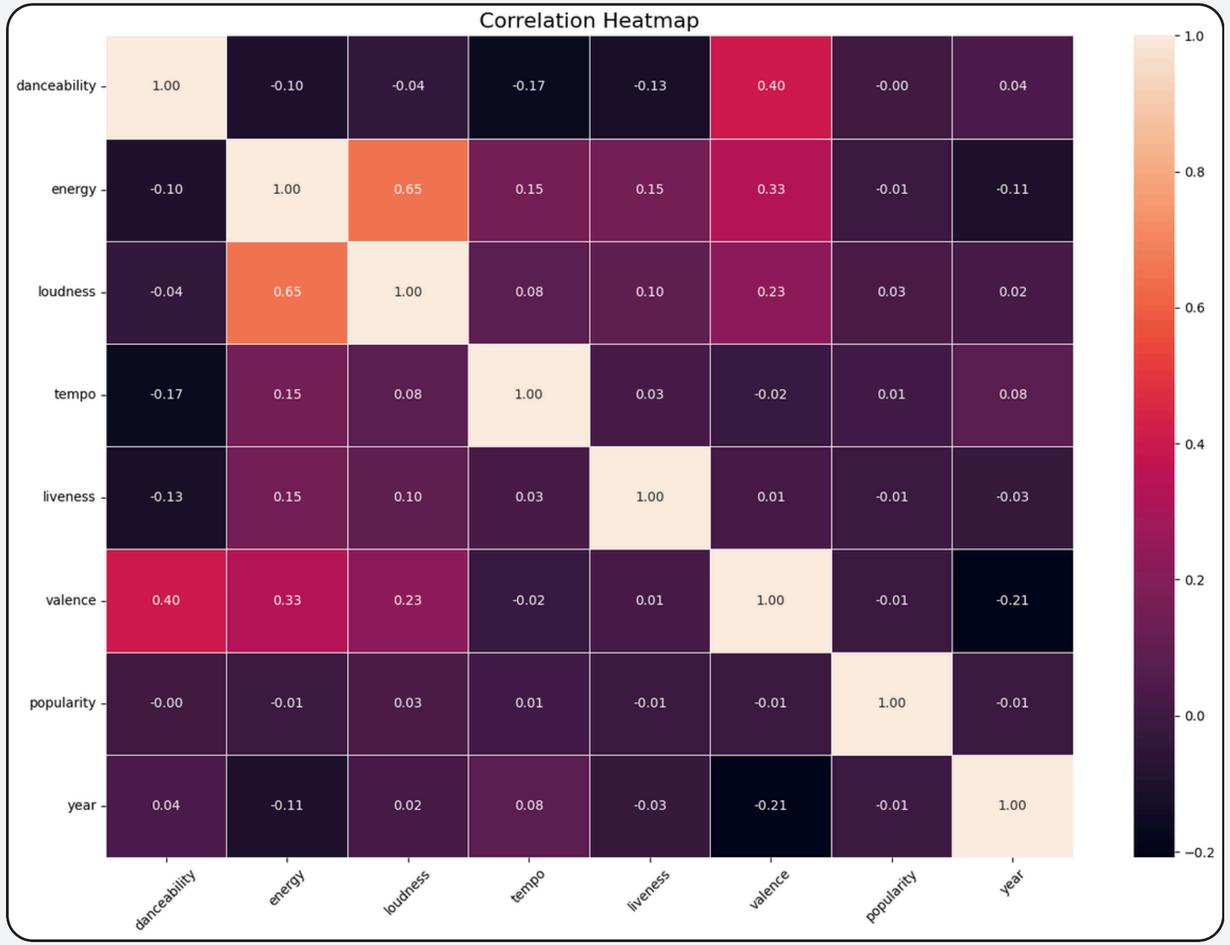
Data Sources

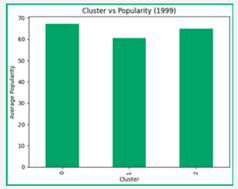
Datasets	Type	Source	Features	Size
Top Hits Spotify [2000-2019]	CSV	Kaggle	artist, song, duration, popularity, year, danceability, energy, loudness, tempo, genre, liveliness, valence	~2000
Spotify API	API	Spotify Developer API	song name, artist, release date, album name	~1000
Billboard Hot 100 Songs	CSV	Kaggle	date, rank, artist, song	~1000

Exploratory Data Analysis & Feature Analysis

- *No individual features are strong enough to explain the popularity of the music.*
- *Strong relationships can be observed between features but not with popularity*
- *Popularity is not a simple, linear outcome*



Correlation Heatmap



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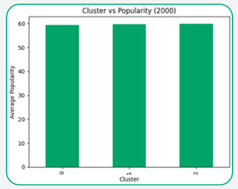

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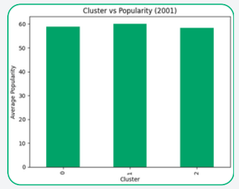

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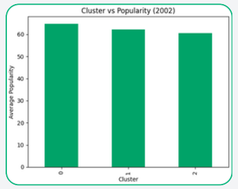

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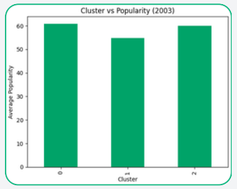

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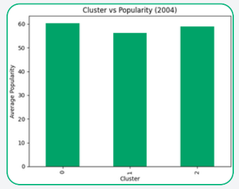

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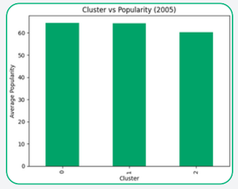

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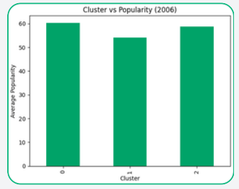

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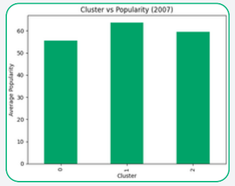

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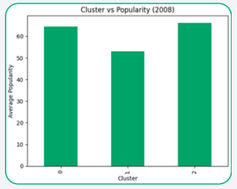

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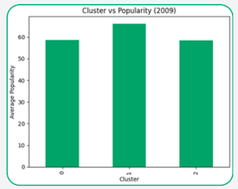

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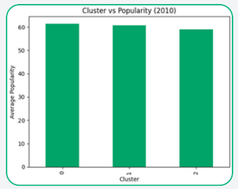

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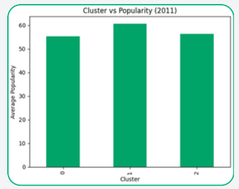

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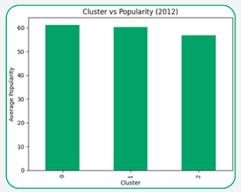

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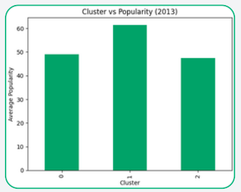

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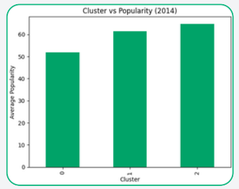

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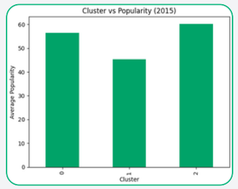

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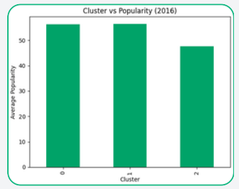

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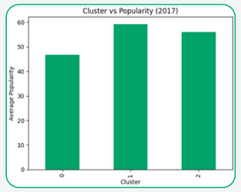

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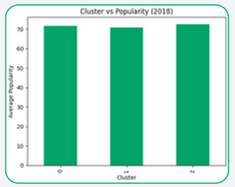

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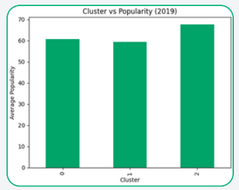


2019

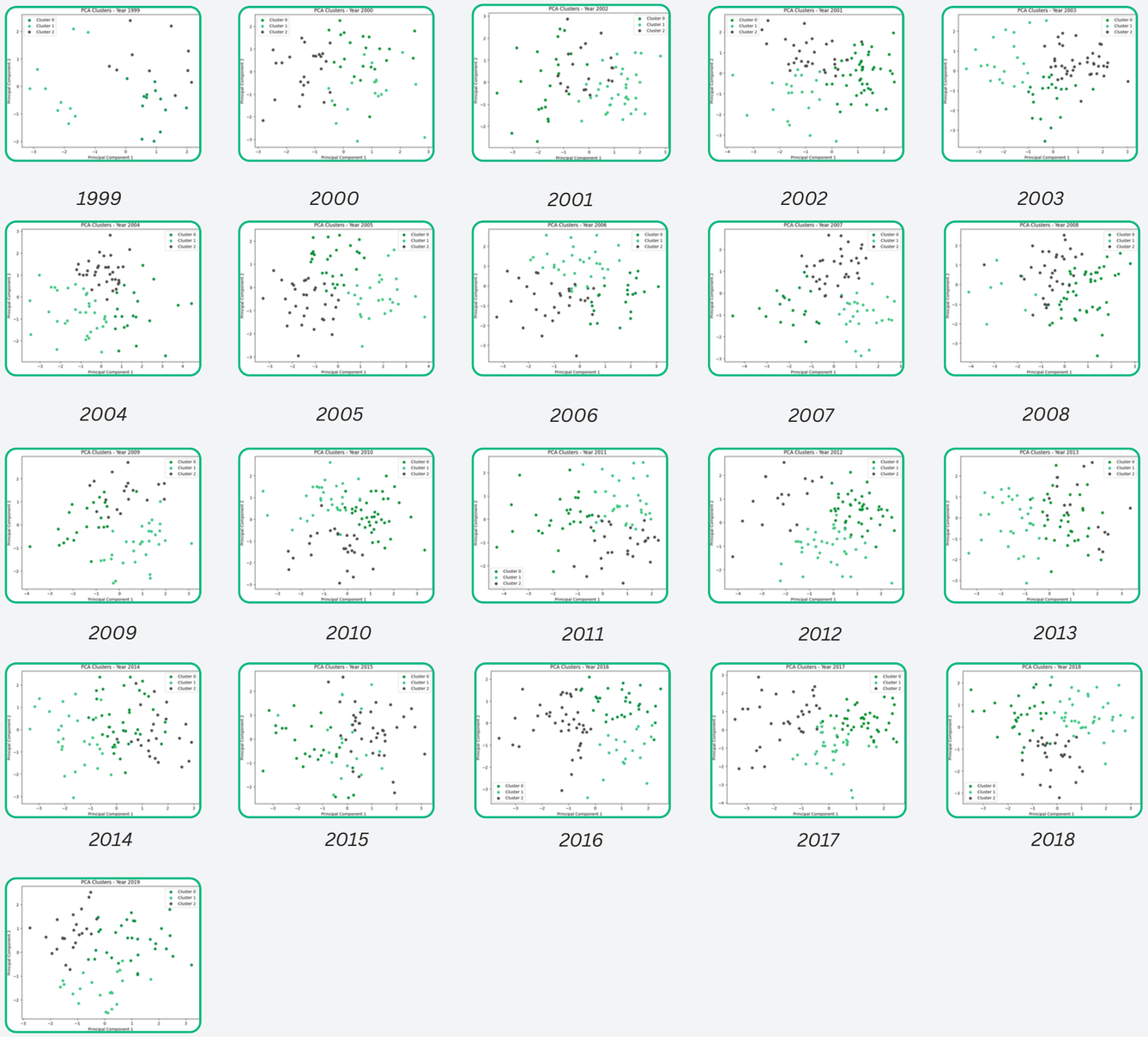
Distribution of Songs across Clusters for each year

K-Means Clustering

- *Songs were grouped into 3 clusters based on their audio features*
  - ***Cluster 0 [High Energy]:*** *Loud and Intense Songs*
  - ***Cluster 1 [Balanced]:*** *Moderate Across All Features*
  - ***Cluster 2 [Emotional]:*** *Softer and More Expressive Songs*
- *No single cluster dominated in popularity — hits came from all 3 groups*

Principal Component Analysis

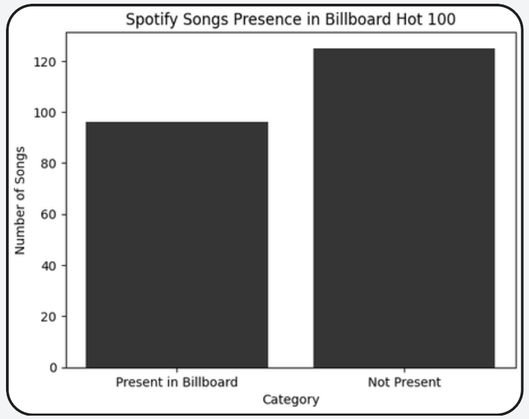
- *Reduced the 6 features into 2 dimensions to visualize the clusters.*
- *Music can be simplified into:*
  - ***PC1 [Intensity]:*** *Driven by Energy and Loudness*
  - ***PC2 [Emotion/Rhythm]:*** *Driven by Valence and Danceability*
- *All 3 clusters separated clearly in the PCA plot*
- *Confirms that our clustering results were valid and meaningful*



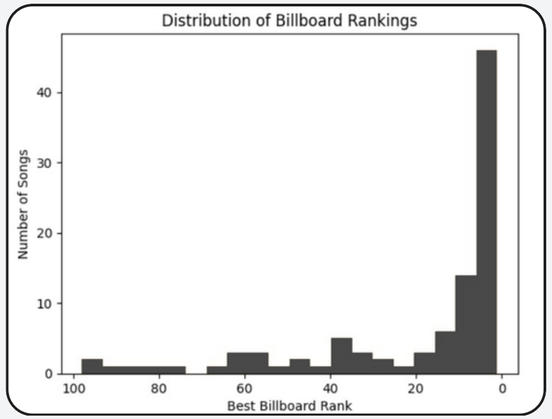
PCA Visualization of Song Clusters

Spotify *vs* Billboard Comparison

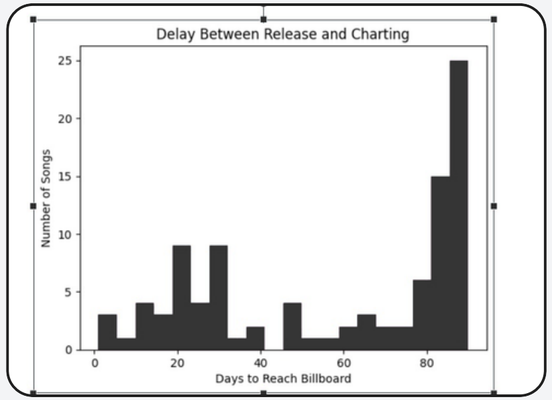
- *57% of Spotify popular songs never appeared on Billboard at all.*
- *Songs that made it to Billboard mostly reached the top 10 - getting there is the real challenge.*
- *Delay before charting varies hugely - some songs chart within days, others take up to 90 days.*
- *There is no consistent timing pattern - chart success is unpredictable.*
- *Spotify popularity and Billboard success measure two completely different things.*



Spotify vs Billboard (Present vs Not Present)



Distribution of Billboard Rankings



Spotify vs Billboard (Charting Time Delay)

Challenges Faced

- *Matching Songs across Datasets (like Name Inconsistencies and Date Formats)*
- *Spotify API Limitations (Rate Limits, Token Expiry)*
- *Date Alignment Issues (Release vs Chart Dates)*
- *Weak Direct Correlations requiring Advanced Methods*

Thank You

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Questions

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